

WebAPI microservices

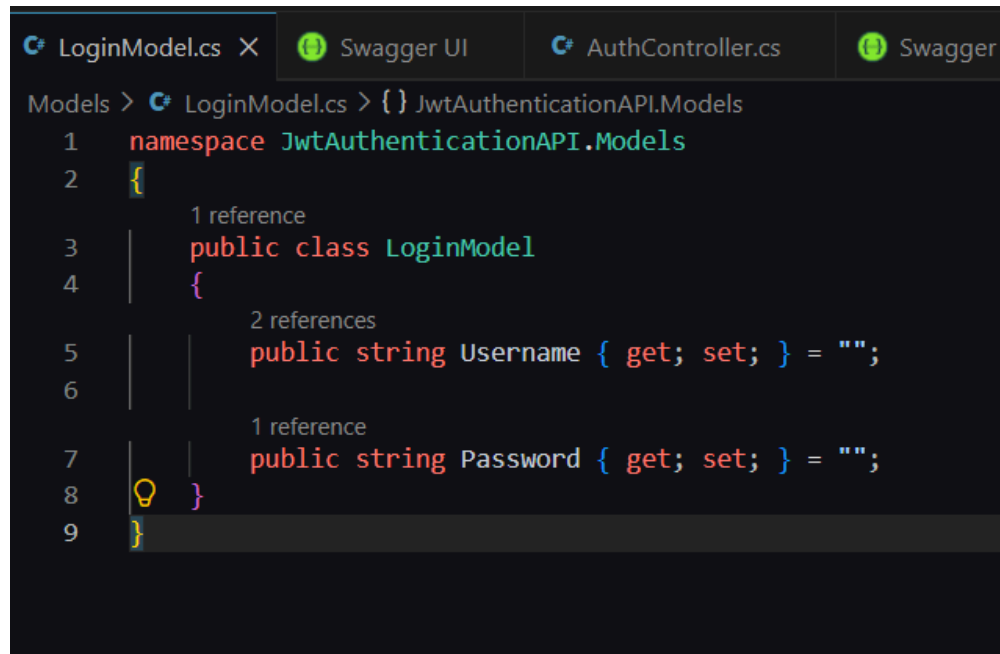
Objective

To implement JWT authentication and authorization in an ASP.NET Core Web API and secure API endpoints using JWT tokens.

1. [Program.cs](#)

```
C# Program.cs
1  using Microsoft.OpenApi.Models;
2  using Microsoft.AspNetCore.Authentication.JwtBearer;
3  using Microsoft.IdentityModel.Tokens;
4  using System.Text;
5  var builder = WebApplication.CreateBuilder(args);
6  // Add Controllers
7  builder.Services.AddControllers();
8  // Swagger
9  builder.Services.AddEndpointsApiExplorer();
10 builder.Services.AddSwaggerGen(c =>
11 {
12     c.SwaggerDoc("v1", new OpenApiInfo
13     {
14         Title = "JWT Authentication API",
15         Version = "v1"
16     });
17     c.AddSecurityDefinition("Bearer", new OpenApiSecurityScheme
18     {
19         Name = "Authorization",
20         Type = SecuritySchemeType.Http,
21         Scheme = "Bearer",
22         BearerFormat = "JWT",
23         In = ParameterLocation.Header,
24         Description = "Enter JWT Token"
25     });
26     c.AddSecurityRequirement(new OpenApiSecurityRequirement
27     {
28         {
29             new OpenApiSecurityScheme
30             {
31                 Reference = new OpenApiReference
32                 {
33                     Type = ReferenceType.SecurityScheme,
34                     Id = "Bearer"
35                 }
36             },
37             Array.Empty<string>()
```

[LoginModel.cs:](#)



The screenshot shows a Visual Studio code editor with four tabs at the top: 'LoginModel.cs' (active), 'Swagger UI', 'AuthController.cs', and another 'Swagger UI'. The 'LoginModel.cs' tab displays the following C# code:

```
Models > LoginModel.cs > { } JwtAuthenticationAPI.Models
1 namespace JwtAuthenticationAPI.Models
2 {
3     1 reference
4     public class LoginModel
5     {
6         2 references
7         public string Username { get; set; } = "";
8         1 reference
9         public string Password { get; set; } = "";
10    }
11 }
```

The code defines a namespace 'JwtAuthenticationAPI.Models' containing a 'LoginModel' class. The class has two public string properties: 'Username' and 'Password', both with get and set accessors and initialized to an empty string. The editor shows line numbers 1 through 9. There are also reference counts shown next to the class and property declarations.

[AuthController.cs:](#)

```

LoginModel.cs | Swagger UI | AuthController.cs | Swagger UI | Program.cs | Swagger UI
controllers > AuthController.cs > {} JwtAuthenticationAPI.Controllers > AuthController > GenerateJwtToken(string)
8 namespace JwtAuthenticationAPI.Controllers
12 public class AuthController : ControllerBase
15 public IActionResult Login([FromBody] LoginModel model)
23 {
24     // token = token
25     // ...
26 }
27 return Unauthorized("Invalid Username or Password");
28 }
29
1 reference
30 private string GenerateJwtToken(string username)
31 {
32     var claims = new[]
33     {
34         new Claim(ClaimTypes.Name, username)
35     };
36
37     var key = new SymmetricSecurityKey(
38         Encoding.UTF8.GetBytes("ThisIsASecretKeyForJwtToken12345"));
39
40     var credentials = new SigningCredentials(
41         key,
42         SecurityAlgorithms.HmacSha256);
43
44     var token = new JwtSecurityToken(
45         issuer: "MyAuthServer",
46         audience: "MyApiUsers",
47         claims: claims,
48         expires: DateTime.Now.AddMinutes(60),
49         signingCredentials: credentials);
50
51     return new JwtSecurityTokenHandler().WriteToken(token);
52 }
53 }
54 }

```

<http://localhost:5047/swagger/index.html>


Swagger
Supported by SMARTBEAR

Select a definition

JwtAuthenticationAPI v1
 

JWT Authentication API ^{v1} OAS 3.0

<http://localhost:5047/swagger/v1/swagger.json>

Authorize 

Auth 

POST /api/Auth/login  

Curl

```
curl -X 'POST' \
  'http://localhost:5047/api/Auth/login' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "username": "admin",
    "password": "admin123"
  }'
```

Request URL

http://localhost:5047/api/Auth/login

Server response

Code	Details
500	Error: Internal Server Error

Undocumented

Response body

```
at Microsoft.IdentityModel.Tokens.JwtTokenUtilities.CreateEncodedSignature(String input, SigningCredentials signingC
at Microsoft.IdentityModel.Tokens.JwtTokenUtilities.CreateEncodedSignature(String input, SigningCredentials signingC
at System.IdentityModel.Tokens.Jwt.JwtSecurityTokenHandler.WriteToken(SecurityToken token)
at JwtAuthenticationAPI.Controllers.AuthController.GenerateJwtToken(String username) in C:\Users\suman\OneDrive\Desktop\JwtAuthentic
ntroller.cs:line 51
at JwtAuthenticationAPI.Controllers.AuthController.Login(LoginModel model) in C:\Users\suman\OneDrive\Desktop\JwtAuthentic
```

JWTAUTHENTICATIONAPI

- bin
- Controllers
 - AuthController.cs
- Models
 - LoginModel.cs
- obj
- Properties
- appsettings.Development.js...
- appsettings.json
- JwtAuthenticationAPI.csproj
- JwtAuthenticationAPI.http
- Program.cs

http://localhost:5047/swagger/index.html

Swagger

Supported by SMARTBEAR

Select a definition

JwtAuthenticationAPI v1

Available authorizations

Bearer (http, Bearer)

Authorized

Enter JWT Token

Value: *****

Logout Close

Auth